Distributed Systems Project 2

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System Functionality:

The system has 3 main components (the supernode, the compute node(s), and the client). The supernode is the first to start, and acts as an entry point for compute nodes requesting to join the network. It handles them by communicating with the existing network to see if the node can join (also utilizing a pseudo-firewall via its *compute_nodes.txt* file to determine which endpoints are allowed to join the compute network). This also enables nodes to dynamically join the network (even as training tasks are actively occurring), where they can subsequently assist in workflow contribution.

Clients attempt to connect to the supernode, which helps to broker a connection between the client and an 'entry point' within the network ring. Then, the client will send its training files out to the node network, and create its model by averaging their results, and uses that model to calculate and report the validation error on its validation file.

As previously mentioned, each compute node will either begin the network (in the case of a node joining an empty network) or join in. Each node maintains a finger table, which indicates which node & address a hashed key for a request should be routed to. To simplify things, each node keeps the addresses of its predecessor and successor node. When a node joins or leaves the network, nodes use their connections to update their finger tables (and thus the records of where to route tasks based on their hashed keys). Finally, each node keeps data and a list of files its trained.

All of these components work together to effectively formulate a DHT (distributed hash table) based federated learning system, which drastically improves efficiency.

SMALL RUN (1 node):

Supernode: csel-kh1250-06.cselabs.umn.edu,9090
Node 1: csel-kh1250-06.cselabs.umn.edu,8091
Client: connects to 9090

Node 1 (8091) Init	Supernode Output

```
knick073@csel-kh1250-06:~/csci5105/PA2$ python3 compute_node.py 8091
Starting compute node server on port 8091...
Received node ID: 5
Current network size: 0
This is the first node in the network
Initialized finger table with 4 entries

---- Node Information ----
Node ID: 5
IP:Port: 134.84.182.206:8091
Predecessor: ID=5, IP:Port=134.84.182.206:8091
Successor: ID=5, IP:Port=134.84.182.206:8091

---- Finger Table ----
Finger 1: Node=5, Range=[6, 7)
Finger 2: Node=5, Range=[7, 9)
Finger 3: Node=5, Range=[9, 3)
Finger 4: Node=5, Range=[3, 1)

---- Stored Data ----
Accepted key range: [0, 1, 2, 3, 4, 5]
Stored files: []
Trained files: []
```

Node 1 Output (during training)

```
knick073@csel-kh1250-06:~/csci5105/PA2$ python3 supernode.py
Loaded 12 node addresses
Supernode initialized with max_nodes = 10
Starting the supernode server...
Node with ip:port csel-kh1250-06.cselabs.umn.edu:8091 is requesting to join with ID 5
Empty network, no nodes available
Node with ID 5 confirmed join
Current nodes: {5: NodeAddress(ip='csel-kh1250-06.cselabs.umn.edu', port=8091, id=5)}
Returning random node with ID 5
```

Client Output (post-training)

```
Node 5: Checking local model for file rbTKPSgtoD
Node 5: File rbTKPSgtoD is still training
Node 5: Received get_model for file pGzTV7h0GV with key 1
Node 5: Checking local model for file pGzTV7h0GV
Node 5: File pGzTV7h0GV is still training
Node 5: Received get_model for file s0DTFqdTuV with key 9
Node 5: Checking local model for file s0DTFqdTuV
Node 5: File s0DTFqdTuV is still training
Node 5: Received get_model for file yfLfi9XxhP with key 0
Node 5: Checking local model for file yfLfi9XxhP
Node 5: File yfLfi9XxhP is still training
Node 5: Received get_model for file CDEPLr8yIh with key 2
Node 5: Checking local model for file CDEPLr8yIh
Node 5: File CDEPLr8yIh is still training
Node 5: Received get_model for file eunqO3YzwJ with key 9
Node 5: Checking local model for file eunqO3YzwJ
Node 5: File eunqO3YzwJ is still training
Node 5: Training complete for file zx0uRKUQj5 with error 0.236
```

```
Checking model for file eunqO3YzwJ
Model for file eunqO3YzwJ is not trained yet
Waiting for training to complete...
Checking model for file eunqO3YzwJ
Model for file eunqO3YzwJ is not trained yet
Waiting for training to complete...
Checking model for file eunqO3YzwJ
Model for file eunqO3YzwJ is not trained yet
Waiting for training to complete...
Checking model for file eunqO3YzwJ
Model for file eunqO3YzwJ is trained
33 models aggregated
Validation error: 0.35942857142857143
Time taken: 344.01073932647705
knick073@csel-kh1250-06:~/csci5105/PA2$
```

Filepath example 1

```
routing > 2aM2VBjKp1_routing.txt
1   Node ID: 5, Filename: 2aM2VBjKp1, Key: 9, Time: 1743294246.6329894
2
```

Filepath example 2

```
routing > 4H3fjyoCmg_routing.txt
1   Node ID: 5, Filename: 4H3fjyoCmg, Key: 9, Time: 1743294246.6573482
2
```

REGULAR RUN (4 nodes):

Ran on nodes (in-order of initiation):

Supernode: csel-kh1250-06.cselabs.umn.edu,9090
Node 1: csel-kh1250-06.cselabs.umn.edu,8091
Node 2: csel-kh1250-06.cselabs.umn.edu,8092
Node 3: csel-kh1250-06.cselabs.umn.edu,8093
Node 4: csel-kh1250-06.cselabs.umn.edu,8094
Client: connects to 9090

NODE OUTPUTS

- From POV of first node (8091). Nodes joined in ascending order wrt. their port #s here:

Init

```
knick073@csel-kh1250-06:~/csci5105/PA2$ python3 compute_node.py 8091
Starting compute node server on port 8091...
Received node ID: 7
Current network size: 0
This is the first node in the network
Initialized finger table with 4 entries

---- Node Information ----
Node ID: 7
IP:Port: 134.84.182.206:8091
Predecessor: ID=7, IP:Port=134.84.182.206:8091
Successor: ID=7, IP:Port=134.84.182.206:8091

---- Finger Table ----
Finger 1: Node=7, Range=[8, 9)
Finger 2: Node=7, Range=[9, 1)
Finger 3: Node=7, Range=[1, 5)
Finger 4: Node=7, Range=[5, 3)

---- Stored Data ----
Accepted key range: [0, 1, 2, 3, 4, 5, 6, 7]
Stored files: []
Trained files: []
```

8092 joins

```
---- Node Information ----
Node ID: 7
IP:Port: 134.84.182.206:8091
Predecessor: ID=9, IP:Port=134.84.182.206:8092
Successor: ID=9, IP:Port=134.84.182.206:8092

---- Finger Table ----
Finger 1: Node=9, Range=[8, 9)
Finger 2: Node=9, Range=[9, 1)
Finger 3: Node=9, Range=[1, 5)
Finger 4: Node=9, Range=[5, 3)

---- Stored Data ----
Accepted key range: [0, 1, 2, 3, 4, 5, 6, 7, 0, 1, 2, 3, 4, 5, 6, 7]
Stored files: []
Trained files: []
Successor 134.84.182.206:8092 is reachable!
```

8093 joins

```
---- Node Information ----
Node ID: 7
IP:Port: 134.84.182.206:8091
Predecessor: ID=4, IP:Port=134.84.182.206:8093
Successor: ID=9, IP:Port=134.84.182.206:8092

---- Finger Table ----
Finger 1: Node=9, Range=[8, 9)
Finger 2: Node=9, Range=[9, 1)
Finger 3: Node=4, Range=[1, 5)
Finger 4: Node=9, Range=[5, 3)

---- Stored Data ----
Accepted key range: [5, 6, 7]
Stored files: []
Trained files: []
Successor 134.84.182.206:8092 is reachable!
```

8094 joins

```
---- Node Information ----
Node ID: 7
IP:Port: 134.84.182.206:8091
Predecessor: ID=4, IP:Port=134.84.182.206:8093
Successor: ID=9, IP:Port=134.84.182.206:8092

---- Finger Table ----
Finger 1: Node=9, Range=[8, 9)
Finger 2: Node=9, Range=[9, 1)
Finger 3: Node=4, Range=[1, 5)
Finger 4: Node=7, Range=[5, 3)

---- Stored Data ----
Accepted key range: [5, 6, 7]
Stored files: []
Trained files: []
Successor 134.84.182.206:8092 is reachable!
```

Training

```
Node 7: File 44NUCe174R is still training
Node 7: Received get_model for file 4H3fjyoCmg with key 9
Node 7: Forwarding get_model for file 4H3fjyoCmg to node 9
Node 7: Received get_model for file s0DTFqdTuV with key 9
Node 7: Forwarding get_model for file s0DTFqdTuV to node 9
Node 7: Received get_model for file eunqO3YzwJ with key 9
Node 7: Forwarding get_model for file eunqO3YzwJ to node 9
Node 7: Received get_model for file FF1I48s675 with key 7
Node 7: Checking local model for file FF1I48s675
Node 7: File FF1I48s675 is still training
Node 7: Received get_model for file YHCHiXhKWy with key 6
Node 7: Checking local model for file YHCHiXhKWy
Node 7: File YHCHiXhKWy is still training
Node 7: Received get_model for file rM9QAoFYcB with key 5
Node 7: Checking local model for file rM9QAoFYcB
Node 7: File rM9QAoFYcB is still training
Node 7: Received get_model for file YorO5gpN2K with key 5
Node 7: Checking local model for file YorO5gpN2K
Node 7: File YorO5gpN2K is still training
Node 7: Received get_model for file V0msie6NEw with key 6
Node 7: Checking local model for file V0msie6NEw
Node 7: File V0msie6NEw is still training
Node 7: Received get_model for file o6ntpH1q6N with key 5
Node 7: Checking local model for file o6ntpH1q6N
Node 7: File o6ntpH1q6N is still training
```

Post-Training (client output)

```
Model for file CDEPLr8yIh is not trained yet
Waiting for training to complete...
Checking model for file 44NUCe174R
Model for file 44NUCe174R is not trained yet
Checking model for file CDEPLr8yIh
Model for file CDEPLr8yIh is not trained yet
Waiting for training to complete...
Checking model for file 44NUCe174R
Model for file 44NUCe174R is not trained yet
Checking model for file CDEPLr8yIh
Model for file CDEPLr8yIh is trained
Waiting for training to complete...
Checking model for file 44NUCe174R
Model for file 44NUCe174R is trained
33 models aggregated
Validation error: 0.35942857142857143
Time taken: 135.87337017059326
knick073@csel-kh1250-06:~/csci5105/PA2$
```

Node Route Example 1

routing 〉 byKWt8V4pT_routing.txt

```
~/csci5105/PA2/routing
ename: byKWt8V4pT, Key: 3, Time: 1743295661.2123742
Node ID: 4, Filename: byKWt8V4pT, Key: 3, Time: 1743295661.2320096
```

Node Route Example 2

routing 〉 YHCHiXhKWy_routing.txt

```
Node ID: 0, Filename: YHCHiXhKWy, Key: 6, Time: 1743295661.256593
Node ID: 4, Filename: YHCHiXhKWy, Key: 6, Time: 1743295661.305586
Node ID: 7, Filename: YHCHiXhKWy, Key: 6, Time: 1743295661.3398676
```

Node Route Example 3

routing 〉 s0DTFqdTuV_routing.txt

```
Node ID: 0, Filename: s0DTFqdTuV, Key: 9, Time: 1743295663.2167773
Node ID: 4, Filename: s0DTFqdTuV, Key: 9, Time: 1743295663.262087
Node ID: 7, Filename: s0DTFqdTuV, Key: 9, Time: 1743295663.2771208
Node ID: 9, Filename: s0DTFqdTuV, Key: 9, Time: 1743295663.3215454
```

Supernode Output (post-initialization)

```
knick073@csel-kh1250-06:~/csci5105/PA2$ python3 supernode.py
Loaded 12 node addresses
Supernode initialized with max_nodes = 10
Starting the supernode server...
Node with ip:port csel-kh1250-06.cselabs.umn.edu:8091 is requesting to join with ID 7
Empty network, no nodes available
Node with ID 7 confirmed join
Current nodes: {7: NodeAddress(ip='csel-kh1250-06.cselabs.umn.edu', port=8091, id=7)}
Node with ip:port csel-kh1250-06.cselabs.umn.edu:8092 is requesting to join with ID 9
Returning random node with ID 7
Node with ID 9 confirmed join
Current nodes: {7: NodeAddress(ip='csel-kh1250-06.cselabs.umn.edu', port=8091, id=7), 9: NodeAddress(ip='csel-kh1250-06.csel
abs.umn.edu', port=8092, id=9)}
Node with ip:port csel-kh1250-06.cselabs.umn.edu:8093 is requesting to join with ID 4
Returning random node with ID 7
Node with ID 4 confirmed join
Current nodes: {7: NodeAddress(ip='csel-kh1250-06.cselabs.umn.edu', port=8091, id=7), 9: NodeAddress(ip='csel-kh1250-06.csel
abs.umn.edu', port=8092, id=9), 4: NodeAddress(ip='csel-kh1250-06.cselabs.umn.edu', port=8093, id=4)}
Node with ip:port csel-kh1250-06.cselabs.umn.edu:8094 is requesting to join with ID 0
Returning random node with ID 4
Node with ID 0 confirmed join
Current nodes: {7: NodeAddress(ip='csel-kh1250-06.cselabs.umn.edu', port=8091, id=7), 9: NodeAddress(ip='csel-kh1250-06.csel
abs.umn.edu', port=8092, id=9), 4: NodeAddress(ip='csel-kh1250-06.cselabs.umn.edu', port=8093, id=4), 0: NodeAddress(ip='cse
l-kh1250-06.cselabs.umn.edu', port=8094, id=0)}
```

RESULTS DISCUSSION

Our trial runs found that increasing the number of nodes reduced the training time for our 33 files from 344 seconds (one node) to ~136 seconds (4 nodes). We were able to see how the routing table of a node changes as new nodes are added to the network, and multiple examples of how a training task can be routed through the compute nodes within the network. This system reduces training time for distributed learning tasks while requiring low overhead for nodes and does not rely on a centralized scheduler, which yields high reliability and resiliency.